

OMNICATIONER: One Captioner to Rule Them All

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<https://alpha-innovator.github.io/OmniCaptioner-project-page>

<https://github.com/Alpha-Innovator/OmniCaptioner>

<https://huggingface.co/U4R/OmniCaptioner>

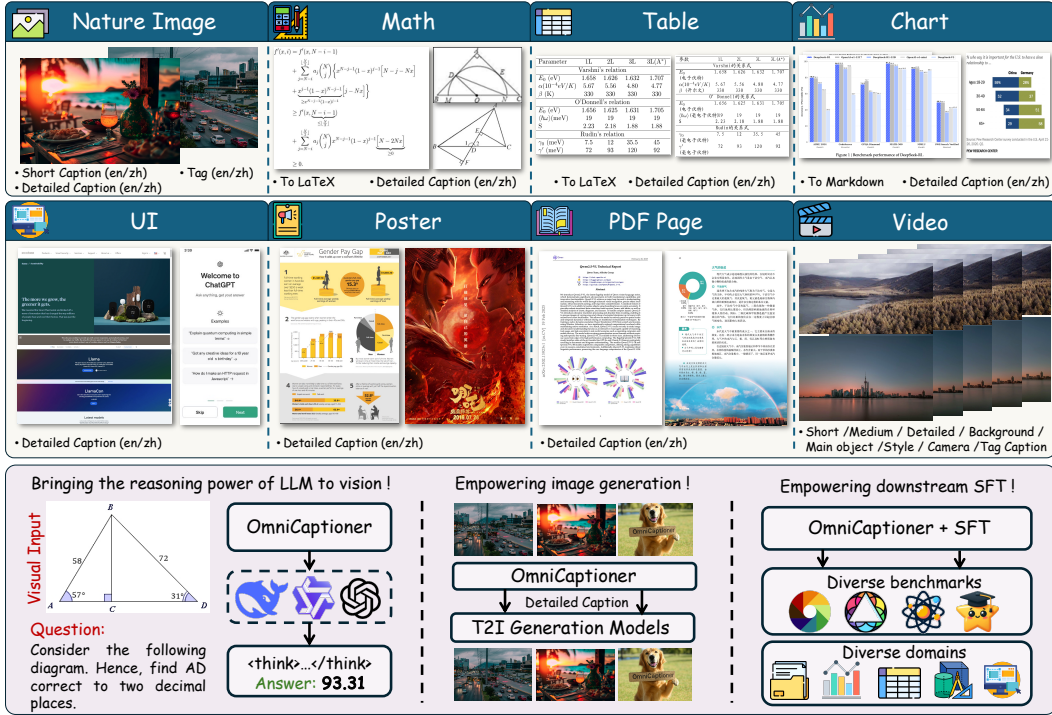


Figure 1: **OMNICATIONER**: the top section demonstrates its capability to process diverse visual domains. The bottom section highlights its applications in visual reasoning (associated with reasoning LLM), image generation (integrated with T2I generation models), and efficient downstream SFT tasks adaptation.

Abstract

We propose OMNICATIONER, a versatile visual captioning framework for generating fine-grained textual descriptions across a wide variety of visual domains. Unlike prior methods limited to specific image types (e.g., natural images or geometric visuals), our framework provides a unified solution for captioning natural

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images, visual text (*e.g.*, posters, UIs, textbooks), and structured visuals (*e.g.*, documents, tables, charts). By converting low-level pixel information into semantically rich textual representations, our framework bridges the gap between visual and textual modalities. Our results highlight three key advantages: (i) Enhanced Visual Reasoning with LLMs, where long-context captions of visual modalities empower LLMs, particularly the DeepSeek-R1 series, to reason effectively in multimodal scenarios; (ii) Improved Image Generation, where detailed captions improve tasks like text-to-image generation and image transformation; and (iii) Efficient Supervised Fine-Tuning (SFT), which enables faster convergence with less data. We believe the versatility and adaptability of OMNICATIONER can offer a new perspective for bridging the gap between language and visual modalities.

1 Introduction

Pretraining of Multimodal Large Language Models (MLLMs) [23, 18, 6, 39, 2], particularly in bridging the gap between visual and textual domains, has gained significant attention in recent years. Substantial progress has been achieved in image captioning and visual question answering, enabling models to serve as universal visual assistants through large-scale Supervised Fine-Tuning (SFT). However, MLLMs still face limitations in perceptual accuracy in the visual-text and structured image domains, particularly when handling synthesized images that exhibit a substantial domain gap from natural images, as illustrated in Fig. 3 (c).

Recent research has increasingly emphasized the role of image captioning in aligning modalities during multimodal pretraining, aiming to enhance both perception and reasoning across diverse domains through the SFT process. Meanwhile, domain-specific studies, such as those focusing on document understanding MLLMs [26, 15] and mathematical MLLMs [33, 54, 41], have leveraged domain-specific caption data to further improve modality alignment and advance multimodal pretraining. These advancements highlight the need for a unified framework for multimodal pretraining centered on image captioning. Also, despite progress in MLLMs, their multimodal reasoning capabilities still fall short of the textual reasoning abilities of LLMs. As shown in Fig. 2, when provided only with a question and no visual input on the MathVision and MathVerse benchmarks, DeepSeek-Distill-Qwen-7B (orange) significantly outperforms Qwen2-VL-Instruct (blue), demonstrating the strength of LLM-driven reasoning in multimodal tasks.

In this work, we bridge this gap by introducing the first OMNICATIONER framework, designed to generate fine-grained textual descriptions across diverse visual domains as shown in Fig. 1. Unlike prior approaches that focus on specific visual categories (*i.e.*, natural or geometry images), our approach enables a unified solution for diverse image types, paving the way for broader multimodal understanding. We focus on converting low-level pixel features into semantically rich textual representations, which preserve crucial visual details while bridging the modality gap between vision and language. OMNICATIONER has two characteristics: i) **Diverse Visual Domain Coverage**: We present a unified framework that supports diverse visual content, including natural images, visual text images (*e.g.*, poster, UI, textbook) and structured images (*e.g.*, geometry, equation, tables, charts). ii) **Pixel-to-Text Mapping**: By pairing these diverse image types with detailed captions, we convert low-level pixel information into semantically rich, fine-grained textual descriptions, enabling a deeper understanding of visual content, which effectively bridges the gap between visual and textual modalities.

To evaluate the effectiveness of OMNICATIONER, we conduct systematic assessments across both image understanding (*e.g.*, visual reasoning) and image generation tasks (*e.g.*, text-to-image generation). Our results reveal several key advantages: i) **Improved Visual Reasoning with LLMs**: Our detailed, long-context captions can be directly incorporated into powerful LLMs to address challenging visual reasoning questions, particularly for models like the DeepSeek-R1 [13] series. This approach enables LLMs to perform visual reasoning tasks in a training-free manner, leveraging rich textual descriptions without requiring additional fine-tuning. ii) **Enhanced Image Generation and Conversion**: The detailed captions produced by our framework significantly improve image generation tasks, such as image-to-text generation and image conversion, owing to their near-complete pixel-to-text mapping capability. iii) **Efficient SFT Process**: Leveraging the knowledge from pretraining on OMNICATIONER, the SFT process becomes more efficient, requiring less training data and achieving faster convergence.

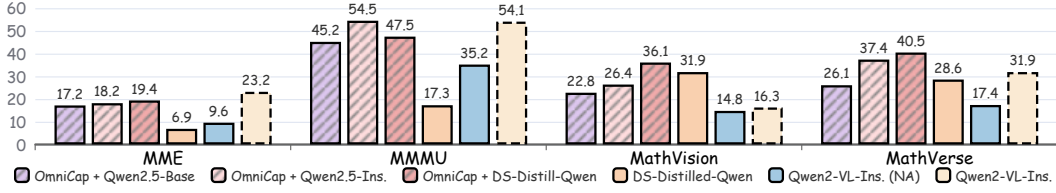


Figure 2: Performance comparison across different visual benchmarks for different LLMs/MLLMs (7B) with or without visual input. The bar with dashed borders denotes Qwen2-VL-Instruct, indicating it has pixel-level visual input, while others do not. Qwen2-VL-Ins.(NA) refers to a setting where only the question is provided as input. We divide the MME score by 100 to have the same scale as other benchmarks.

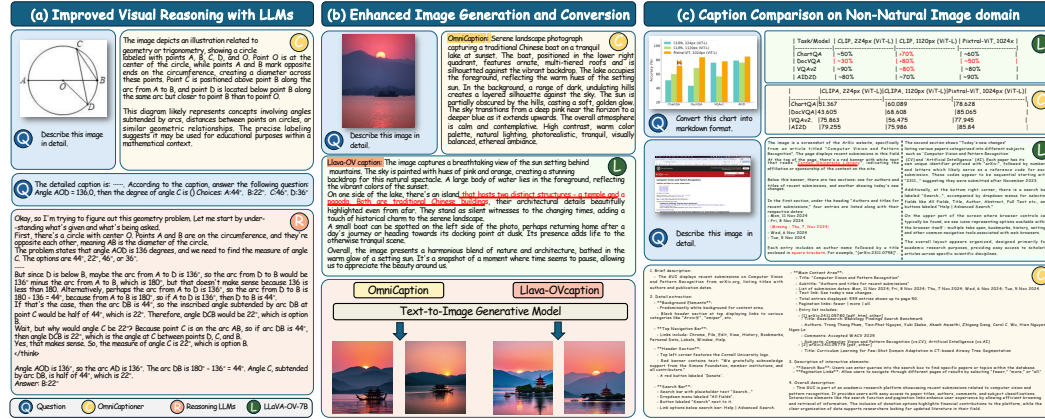


Figure 3: Illustration of OMNICAPTIONER’s plug-and-play applications (Sub-figure a, b) and comparison between OMNICAPTIONER and LLaVA-OneVision-7B on non-natural image captioning (Sub-figure c). Sub-figure (a) shows that OMNICAPTIONER leverages LLMs’ strong reasoning abilities to perform multimodal reasoning tasks. Sub-figure (b) highlights how hallucinated or inaccurate captions—like those from LLaVA-OneVision-7B can lead to inconsistent image conversion, revealing weakened alignment capabilities in text-to-image models when captions don’t faithfully represent the original content. Sub-figure (c) highlights that LLaVA-OneVision-7B, due to limited exposure to non-natural images during pretraining, struggles with perception in such domains, often leading to hallucinations, whereas OMNICAPTIONER provides more accurate descriptions.

Furthermore, the contributions of this paper are summarized below:

- **Unified Visual Captioning Framework:** We present OMNICAPTIONER, a unified framework for generating captions across diverse domains. Our approach seamlessly integrates captioning capabilities for natural images, visual text images (e.g., posters, UI, and text-books), and structured visual images (e.g., tables, charts, equations, and geometric diagrams). OMNICAPTIONER sets a new standard for generalized visual captioning, enabling more effective and scalable vision-language understanding.
- **Comprehensive Pixel-to-Text Conversion:** Our framework leverages detailed captions to convert low-level pixel information into semantically rich, fine-grained textual descriptions, effectively bridging the gap between visual and textual modalities. Particularly, this enhances text-to-image generation by providing more precise and context-aware textual guidance, leading to improved visual fidelity and alignment with the intended semantics.
- **Improved Visual Reasoning with LLMs:** By incorporating detailed, long-context captions, our approach enables enhanced visual reasoning capabilities, especially when integrated into LLMs such as the DeepSeek-R1 series. Leveraging the perceptual information provided by OMNICAPTIONER, LLMs can infer and reason within the textual space to effectively solve visual reasoning tasks.

2 Related Works

Image Captioning. Image captioning tasks can be broadly classified into two categories. The first approach focuses on generating high-quality captions for natural images. Notably, ShareGPT4V [3] improves vision-language alignment by collecting high-quality, attribute-specific captions through targeted prompts to GPT-4V for natural images, while models like Densefusion [19] leverage multiple expert models to synthesize captions for natural images. The second approach, exemplified by CompCap [5], tackles the challenge of domain diversity during pretraining by incorporating synthetic images to enhance performance on underrepresented domains. However, the first approaches are often constrained by its focus on specific domains, while the second faces challenges due to the relatively small quantity of synthetic images used during training.

Multimodal Large Language Models. With the development of LLMs [44, 13, 37, 48], integrating visual perception capability into LLMs (*i.e.*, MLLMs) has received increasing attention. To address the gap between different modalities, most of works [39, 2, 6, 42, 23, 18, 21, 22] first pretrain on image captioning data to obtain a vision-language connector (*e.g.*, MLP-based or cross-attention based) and followed by SFT. To better integrate information from multiple modalities, several works [20, 27, 8, 35] try to explore new architectures to process different modalities in a single Transformer model. In addition to model architecture, some works [40] try to boost models’ reasoning ability through post-training (*e.g.*, reinforcement learning) [40] or test-time scaling (*e.g.*, monte-carlo tree search) [45, 28, 9]. Furthermore, recent studies [50, 31, 5, 7] have systematically investigated the influence of data quality on both the pretraining and SFT phases of MLLMs. MM1 [50] reveals that model capabilities induced through pretraining with high-quality data are effectively preserved after SFT. Most existing open-source MLLMs [23, 18] primarily focus on pretraining with natural images, while domain-specific MLLMs (*e.g.*, math, chart) are trained exclusively on domain-specific caption data. In contrast, we propose a more unified pretraining approach that integrates diverse domain knowledge during pretraining. In addition, current MLLMs generally exhibit inferior reasoning capabilities compared to text-only LLMs, OMNICATIONER can generate detailed, long-context captions of different domains and use LLMs to address challenging visual reasoning tasks.

3 OMNICATIONER

To achieve a unified multimodal pretraining paradigm and handle diverse visual domains, we first construct a diverse caption dataset as shown in Sec. 4. We will provide the dataset description and then detail the dataset construction process in Sec. 3.1 and Sec. 3.2, respectively. And the pertaining process is described in Sec. 3.3.

3.1 Diverse Visual Caption Dataset

The diversity of our visual caption dataset is characterized by two dimensions: domain diversity (diverse data sources) and caption formula diversity. To achieve effective unified pretraining, the dataset needs to encompass a wider range of domains. For example, when acting as a documentation assistant, MLLMs need to comprehend tables and charts, while as a GUI agent, they are required to understand elements in web pages. As illustrated in the data distribution section of Fig. 4, our caption dataset is composed of four major categories: natural images, structured images (including chart, table, and so on), visual text images (including UI images, posters, and so on), and video. This comprehensive data coverage enables our model to serve as a multi-domain assistant and further enhance the performance on downstream tasks. Furthermore, diverse types of captions may be necessary even for the same visual input. For instance, a chart image may require both structured tabular conversion and comprehensive analytical descriptions. To address this requirement, we define diverse caption formulas for each domain. This approach enables our model to generate diverse caption formats, including multilingual (Chinese and English) descriptions, varying granularity levels (from comprehensive to concise), and so on.

3.2 Dataset Construction

To generate high-quality captions for images across diverse domains, we propose a two-step caption generation pipeline. The design of our pipeline takes into account the need for accurate visual

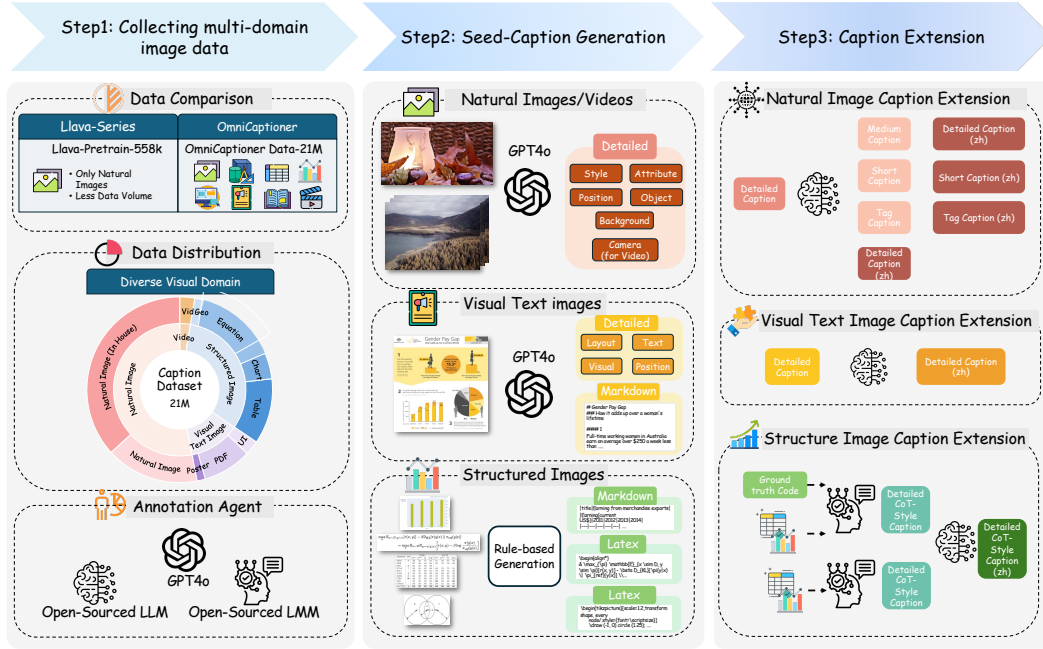


Figure 4: OMNICATIONER’s diverse visual captioning pipeline. The pipeline consists of Seed-Caption Generation to ensure precise pixel-to-word mapping, and Caption Extension to enrich caption styles to support image generation and visual reasoning tasks. OMNICATIONER utilizes a **21M-caption dataset**, covering diverse domains beyond natural images, enabling more comprehensive captioning capabilities. For further details about dataset composition, please refer to Fig. 7 in Appendix A.

descriptions, the flexibility to support different stylistic outputs, the ability to perform reasoning and logic extrapolation, as well as bilingual captioning.

Seed Caption Generation. In the first stage, we focus on seed caption generation. The goal is to produce an initial caption that is as accurate as possible, with a comprehensive textual description of all relevant visual elements present in the visual signal. This stage leverages carefully designed prompts to guide the powerful closed-source multimodal model GPT-4o to describe all possible visual elements in natural images and visual-text images, ensuring an accurate pixel-to-word mapping. For structured images generated via code, the description is generated as accurately as possible using predefined code rules. The generated seed caption serves as a reliable foundation for further refinement in the subsequent stage.

Caption Extension. The second stage, caption extension, is responsible for enhancing and diversifying the generated caption. Here, the focus shifts from purely accuracy to incorporating stylistic variation and domain-specific reasoning. The seed caption is extended by introducing bilingual outputs (Chinese and English), with variations ranging from detailed to medium-length, short, and tag-style captions. Additionally, we inject reasoning knowledge relevant to specific domains to enrich the semantic depth of the captions. This allows the captions to not only reflect the visual content but also accommodate nuanced understanding in different contexts. Specially, for **Natural Images**, we leverage the open-source LLM, Qwen2.5-32B, to adjust the caption length through different prompts, allowing captions to range from medium-length to short and tag-style. Additionally, these varied captions are translated into Chinese, facilitating the creation of bilingual prompts for image generation. The benefit of this approach is to enable more flexible and effective bilingual prompt extraction for image generation tasks. For **Visual Text Images**, we use open source LLM Qwen2.5-32B to translate the detailed subtitles generated by GPT-4o into the corresponding Chinese versions to ensure cross-language consistency. For **Structured Images**, which often relate to mathematical or document-based reasoning (e.g., Chain-of-Thought (CoT) analysis), we prioritize the accuracy of the seed caption. After confirming the seed caption’s accuracy, we input both the seed caption and the original image into the open-source multimodal model Qwen2-VL-76B for CoT-style caption

generation. This approach allows us to condition the captioning process on both the seed caption’s code (*e.g.*, Markdown, LaTeX) and the image content, reducing hallucinations and improving the reliability of the generated captions. Additionally, we collect structured images without seed captions and directly input them into the same multimodal model for CoT-style caption generation. By decoupling the caption generation process into these two stages, we ensure both high accuracy in representing visual content and flexibility in producing diverse, contextually appropriate captions.

3.3 Unified Pretraining Process

To effectively handle the multi-domain nature of the OMNICATIONER dataset, which spans a broad range of image types and captioning tasks, we propose a practical approach utilizing distinct system prompts. These prompts help minimize task conflicts and improve task coordination during training. By customizing system prompts for specific image categories and using a fixed set of question templates for various captioning styles, we differentiate between tasks and data types in the pretraining process. This approach facilitates efficient multi-domain training, ensuring robust model performance across diverse tasks and domains. To address the challenge of handling images with large variations in resolution and arbitrary aspect ratios, we leverage the powerful visual understanding capabilities of the Qwen2-VL-7B [39] model. Given that the Qwen2-VL-Instruct model is inherently powerful in managing multi-domain image data, we initialize our model with the Qwen2-VL-Instruct weights. This initialization allows us to effectively fine-tune on the OMNICATIONER dataset, ensuring robust performance across a wide range of image resolutions and aspect ratios while benefiting from the model’s ability to generalize across diverse domains.

4 One Captioner to Rule Them All

Improved Visual Reasoning Tasks with LLMs. Current MLLMs lag behind LLMs in reasoning capabilities. This discrepancy motivates us to investigate whether LLMs can directly perform visual reasoning without modality-alignment losses that may degrade their reasoning ability while still effectively handling diverse visual reasoning tasks. In this work, we integrate image captioning with large language models (LLMs) to enable seamless visual reasoning in textual space. As illustrated in Fig. 3 (a), firstly, our captioner converts input images (spanning natural images, charts, equations, and beyond) into linguistically dense descriptions that explicitly encode pixel-level structures (*e.g.*, spatial layouts, symbolic operators, tabular hierarchies) into textual space. These captions, acting as lossless semantic proxies, are then directly processed by powerful LLMs (*e.g.*, DeepSeek-R1 [13], Qwen2.5 series [44]) to perform task-agnostic visual reasoning, including geometric problem-solving and spatial analysis.

Just shown in Fig. 3, OMNICATIONER can transform geometric images into detailed and precise visual descriptions. OMNICATIONER accurately describes geometric images, such as a circle with a diameter and circumferential angles, detailing spatial relationships among points. This enables LLMs to perform logical inferences, like calculating angles, without direct pixel-level perception.

There are three key advantages to this approach: i) Decoupled Perception and Reasoning – By separating perception (handled by MLLMs) from reasoning (handled by LLMs), our method avoids conflicts between the two capabilities, leading to more effective and accurate visual reasoning. ii) Elimination of Modality-Alignment Training – Instead of requiring complex modality-alignment losses, our approach translates visual inputs into linguistic representations, allowing LLMs to process them naturally. This removes the need for additional multimodal training while preserving the reasoning strengths of LLMs. iii) Flexibility and Generalization – The plug-and-play design enables seamless integration of LLMs into diverse visual reasoning tasks without domain-specific tuning. This ensures broad applicability across different types of visual inputs, from geometric diagrams to complex tabular structures.

Enhanced Image Generation and Conversion. Detailed and accurate image captions play a pivotal role in both the training and inference stages of Text-to-Image (T2I) tasks. During training, such captions offer fine-grained supervision by explicitly aligning low-level/high-level visual patterns (*e.g.*, textures, spatial arrangements, object attributes) with precise linguistic semantics. At inference time, as shown in Fig. 3(b), detailed and precise captions substantially enhance image generation quality by guiding the model to follow instructions more faithfully—capturing spatial relationships, object

Table 1: Performance comparison on various visual benchmarks between our OMNICATIONER-inserted LLMs and previous SOTA MLLMs. **We would like to emphasize** that by utilizing OMNICATIONER, LLMs can function as MLLMs without requiring additional training. Moreover, we have observed that, particularly in mathematical scenarios, caption-integrated LLMs surpasses MLLMs with comparable parameter sizes, where MLLMs have undergone rigorous data preparation and GPU-intensive training.

Model	MME	MMMU	MathVision	MathVerse	Olympiad
<i>Frontier Models</i>					
GPT-4V	-	63.1	24.0	32.8	18.0
GPT-4o (2024-05)	-	69.1	30.4	50.2	25.9
Claude3.5-Sonnet	-	68.3	-	-	-
<i>3B-Level Models</i>					
Qwen2-VL-2B [39]	1872	41.1	12.4	21.0	-
InternVL2-2B [6]	1876	36.3	12.1	25.3	0.4
MinniCPM-V2.0 [46]	1808	38.2	-	-	-
OMNICATIONER + Qwen2.5-3B-Instruct	1599	43.0	16.0	22.2	7.24
<i>7B-Level Models</i>					
Qwen2-VL-7B [39]	2327	54.1	16.3	31.9	-
InternVL2-8B [6]	2210	52.6	18.4	37.0	1.9
MiniCPM-Llama-V-2.5-8B [47] 2024	45.8	-	-	-	-
Cambrain-1-8B [36]	-	42.7	-	-	-
LLava-Onevision-7B [18]	1998	48.8	-	26.2	-
MiniCPM-V2.6 [47]	2348	49.8	18.3	25.7	-
OMNICATIONER + Qwen2.5-7B-Instruct	1824	54.5	26.4	37.4	10.9
OMNICATIONER + DS-R1-Distill-Qwen-7B	1942	47.5	36.2	40.5	7.8
<i>32B-Level Models</i>					
InternVL-Chat-V1.5 [6]	2194	46.8	15.0	28.4	0.6
InternVL2-26B [6]	2260	51.2	17.0	31.1	3.5
Cambrian-34B [36]	-	49.7	-	-	-
VILA-1.5-40B	-	55.1	-	-	-
InternVL2-40B	2307	55.2	16.9	36.3	3.9
OMNICATIONER + Qwen2.5-32B-Instruct	1831	59.7	32.1	39.7	13.1
OMNICATIONER + DS-R1-Distill-Qwen-32B	2007	59.2	43.3	43.7	13.2
<i>72B-Level Models</i>					
Qwen2-VL-72B [39]	2482	64.5	25.9	-	11.2
InternVL2-76B [6]	2414	62.7	23.6	42.8	5.5
LLaVA-OneVision-72B [18]	2261	56.8	-	39.1	-
OMNICATIONER + DS-R1-Distill-Llama-70B	2025	64.6	42.9	42.5	13.7

interactions, and semantic details with higher fidelity. These benefits highlight the critical role of captions as a dense supervisory signal, enabling more precise instruction-following in T2I generation.

Efficient SFT Process. The training paradigm of MLLMs typically consists of two sequential phases: pretraining on image-caption data, followed by Supervised Fine-Tuning (SFT). Empirical studies [5, 17, 31] have demonstrated that diverse and high-quality image-caption data (*e.g.*, composite images) can significantly enhance image-language alignment and subsequently promote performance on downstream tasks, such as Visual Question Answering (VQA). OMNICATIONER leverages diverse and high-quality domain data (*e.g.*, table, chart, and so on) during the pretraining phase, enabling the model to acquire multi-domain knowledge. During the SFT phase, the multi-domain knowledge serves as a crucial foundation for rapid adaptation to downstream tasks across different domains.

5 Experiment

To evaluate OMNICATIONER, we conduct three primary experiments. The **first** experiment focuses on visual reasoning with a Caption-inserted Large Language Model. In this setup, detailed captions and corresponding questions are provided to the LLM, and its ability to answer the questions is evaluated. We use five benchmark datasets to assess the model’s performance on this downstream task: MME [10], Mathverse [52], Mathvision [38], MMMU [49] and Olympiad bench [14]. For the LLMs, we select Qwen2.5-3B-Instruct [44], Qwen2.5-7B-Instruct [44], Qwen2.5-

Table 2: Performance comparison of models trained with different captioners on GenEval [12] (Resolution: 1024×1024).

Methods	GenEval $\uparrow$					
	Color Attri.	Sin. Obj.	Pos.	Colors	Counting	Overall
SANA-1.0-1.6B [43]	38.50	98.75	21.25	86.70	65.31	64.61
SANA-1.0-1.6B + Qwen2-VL [39]	44.29	98.44	26.64	86.97	57.81	65.27
SANA-1.0-1.6B + OMNICATIONER	46.00	99.06	29.50	84.57	64.06	67.58

32B-Instruct [44], DeepSeek-R1-Distill-Qwen-7B [13], DeepSeek-R1-Distill-Qwen-32B [13], and DeepSeek-R1-Distill-LLaMA-70B [13], all chosen for their strong reasoning capabilities. The **second** experiment evaluates the efficiency of the SFT process. For this, we select the LLaVA-OneVision [18] data from the OV stage with chain-of-thought enhancement to assess the SFT version of OMNICATIONER across multiple commonly-used benchmarks [10, 49, 29, 30, 38, 52, 25]. The **third** experiment involves finetuning the text-to-image generation model [34, 11, 43] such as SANA-1.0-1.6B [43] with image-caption pairs generated by different captioners (*i.e.*, Qwen2-VL [39], OMNICATIONER). The training setting uses a resolution of 1024×1024 . The model’s generative performance is then evaluated on the GenEval [12].

5.1 Main Results

Improved Visual Reasoning with LLMs. Our experimental results demonstrate that integrating captions into reasoning-enhanced Large Language Models (LLMs), without any additional fine-tuning, achieves state-of-the-art performance across multiple reasoning benchmarks, including MathVision [38], MathVerse [52], MMMU [49], and Olympiad bench [14]. This highlights the power of OMNICATIONER in boosting reasoning capabilities for multiple visual tasks. Specifically, OMNICATIONER-inserted LLMs significantly outperform existing models in MathVision across multiple model sizes, underscoring the enhancement of reasoning ability for complex visual and mathematical tasks. Notably, *OmniCaptioner + DS-R1-Distill-Qwen-7B* and *OmniCaptioner + DS-Distill-Qwen-32B* demonstrate exceptional performance on MathVerse benchmark, significantly outperforming previous models. These results further validate the efficacy of caption-based pretraining in bridging the LLM’s comprehension of visual geometry content. In the MMMU benchmark, *OmniCaptioner + DS-R1-Distill-Qwen-72B* approaches the performance of Qwen2-VL-72B, with a minimal gap between them. This result serves as strong evidence that caption integration with reasoning-enhanced LLMs leads to significant visual understanding and reasoning for multidisciplinary content.

The successful integration of captions with LLMs across scales, from 3B to 72B, underscores that OMNICATIONER consistently enhances LLMs’ reasoning abilities for visual tasks, yielding improvements irrespective of model size. These results highlight that our unified pretraining methodology, leveraging large-scale caption data, is a highly effective strategy for advancing visual reasoning across diverse tasks, outperforming existing approaches even when compared to large-scale fine-tuning methods.

Enhanced Image Generation. As illustrated in Tab. 2, to validate the importance of caption accuracy in T2I generation, our model demonstrates significant performance improvements over the Qwen2-VL-Instruct [39] caption and original SANA, on GenEval benchmark. The original SANA model achieves a 64.61 overall score on GenEval, which is significantly improved to 65.27 with Qwen2-VL-Instruct and further to 67.58 with OMNICATIONER. This +2.97 absolute gain over the vanilla SANA model underscores the effectiveness of high-quality captions in guiding T2I generation. Also, our OMNICATIONER outperforms Qwen2-VL-Instruct across various aspects (except colors), showcasing the enhanced accuracy of our caption generation.

Efficient SFT. In Tab. 3, we compare the performance of several models on visual perception and reasoning tasks, including *LLaVA-OV-7B(SI)*, *LLaVA-OV-7B*, *Qwen2-VL-base+OV SFT*, and our proposed *OmniCaptioner+OV SFT* model. While *LLaVA-OV-7B(SI)* and *LLaVA-OV-7B* use significantly larger datasets for SFT – 3.2M and 4.8M examples, respectively – our *OmniCaptioner+OV SFT* achieves comparable results with just 1.6M SFT examples used during the one-vision (OV) stage. A key difference lies in the unified pretraining phase of OMNICATIONER, which utilizes a diverse caption-based dataset prior to the SFT stage. This step equips the model with richer domain knowledge, enabling it to excel in visual instruction-following tasks despite fewer SFT exam-

Table 3: SFT performance comparison across diverse evaluation benchmarks. OmniCaptioner + OV SFT denotes the SFT model based on OMNICATIONER, while Qwen2-VL-base + OV SFT is based on Qwen2-VL-Base. LLaVA-OV-7B (SI) represents the model after the single-image training in LLaVA-OneVision [18].

SFT Model	Data	MME	MMMU	MathVerse	MathVista	DocVQA	ChartQA
LLaVA-OV-7B (SI) [18]	3.2M	2109	47.3	26.9	56.1	89.3/86.9	78.8
LLaVA-OV-7B [18]	4.8M	1998	48.8	26.2	63.2	90.2/87.5	80.0
Qwen2-VL-Base+OV SFT	1.6M	1905	44.4	24.9	53.8	84.2/-	53.5
OMNICATIONER+OV SFT	1.6M	2045	46.6	25.8	57.4	91.2/-	79.0

Table 4: Comparing different captioners through experiments with captioner-inserted LLM on multiple visual benchmarks.

Captioner Selection	LLM	MME	MMMU	MathVision	MathVerse
llava-onevision-qwen2-7b-ov	DS-R1-Distill-Qwen-7B	1646	22.4	31.7	36.6
InternVL2-8B	DS-R1-Distill-Qwen-7B	1789	23.1	34.4	39.9
Qwen2-VL-7B-Instruct	DS-R1-Distill-Qwen-7B	1914	42.4	31.6	33.0
OMNICATIONER (ours)	DS-R1-Distill-Qwen-7B	1942	47.5	36.2	40.5

ples. It also reveals that *Qwen2-VL-base + SFT* lags behind *OmniCaptioner + OV SFT*, indicating OMNICATIONER’s superior visual perception capabilities.

5.2 Discussions and Findings

We consider conducting three important experiments when combining OMNICATIONER with reasoning-enhanced LLMs. First, we evaluate effectiveness using different Qwen versions. Second, we aim to explore the extent to which Qwen2-VL-Instruct (without image input) and mainstream reasoning-enhanced LLMs rely on visual modality information to solve visual reasoning tasks. Third, we compare OMNICATIONER to Qwen2-VL-Instruct by modifying the captions provided to the reasoning-enhanced LLMs. For more visualization results of image captioning, video captioning, and text-to-image generation task, please refer to Appendix D and Appendix E.

Effect of Different Qwen-Family Versions. Fig. 5 illustrates the performance progression of combining OMNICATIONER with different versions of Qwen on MMMU and MathVerse. As Qwen evolves from Qwen1-8B-chat to Qwen2.5-7B-Instruct, there is a steady improvement in visual reasoning capabilities, driven by the pixel-to-word captioning ability of OMNICATIONER. As illustrated in Fig. 2, the performance comparison between *OmniCaptioner + Qwen2.5-7B-Base*, *OmniCaptioner + Qwen2.5-7B-Instruct* and *OmniCaptioner + DS-R1-Distill-Qwen-7B* highlights the advantage of integrating the DeepSeek Distilled Qwen2.5, which excels in mathematical reasoning. The distilled variant (*DS-R1-Distill-Qwen-7B*) achieves the highest accuracy across *MME (1942)*,

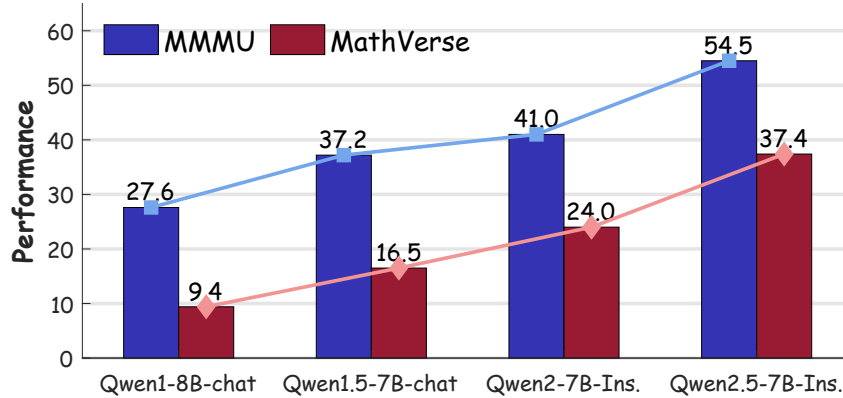


Figure 5: Integrate OMNICATIONER into different versions of LLMs, enabling them to handle tasks in multimodal scenarios.

MathVision (36.2), and *MathVerse* (40.5), emphasizing the benefits of distilled reasoning ability. In contrast, *Qwen2.5-7B-Instruct* is better suited for general world knowledge tasks, as reflected in its improved performance on the *MMMU* (54.5).

Impact of Visual Modality on Reasoning-Enhanced LLMs. From Fig. 2, the performance of *Qwen2-VL-7B (NA)* and *DeepSeek-Distill-Qwen-7B* suggests that the absence of image input significantly restricts their ability to solve visual reasoning tasks. In contrast, *OmniCaptioner + DS-R1-Distill-Qwen-7B*, which retains visual modality, achieves substantially higher accuracy than its non-visual input LLM, highlighting the critical role of visual information in enhancing reasoning capabilities. Furthermore, non-visual input LLM *DS-R1-Distilled-Qwen-7B* significantly outperforms the no-image MLLM (*i.e.*, *Qwen2-VL-Instruct-7B*) on *MathVision* and *MathVerse*, demonstrating the superior reasoning ability of R1 Serious model.

Effect of Different Captioners. Tab. 4 presents a comparative analysis of different captioners on multiple perception and reasoning benchmarks. Our model, incorporating *DeepSeek-Distill-Qwen2.5-7B*, achieves superior performance across all evaluated tasks, significantly outperforming previous approaches. These results highlight the effectiveness of OMNICATIONER, whose captions provide more precise and contextually accurate descriptions than those generated by *Qwen2-VL-7B-Instruct*. The enhanced caption quality contributes to improved visual reasoning tasks, particularly in tasks requiring multi-step inference and detailed visual understanding.

6 Conclusion

We have introduced OMNICATIONER, a unified framework that bridges visual and textual modalities through fine-grained pixel-to-text mapping across diverse domains, including natural images, visual-text images and structured images. By converting low-level visual patterns into semantically rich captions, our approach empowers reasoning-enhanced LLMs (*e.g.*, *DeepSeek-R1*) to achieve enhanced visual reasoning, and enables precise text-to-image generation through comprehensive semantic preservation. This work pioneers a scalable paradigm for multimodal alignment and reasoning, achieving seamless visual-language interoperability without costly label-supervised fine-tuning.

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A OmniCaptioner Dataset Composition

As shown in Fig. 7, the OMNICATIONER dataset is a large-scale multimodal benchmark comprising images, tables, charts, mathematical geometry/equations, posters, PDFs, UI elements, and videos, with captions available in both English and Chinese. The dataset includes natural images sourced from in-house collections, BLIP3Kale [1], and DenseFusion [19]. Tabular data are collected from the arXiv website and the open-source MMTAB dataset [55], while chart data originate from arXiv website and TinyChart [51]. Mathematical content, including equations and geometric structures, is sourced from arXiv and generated from datasets such as MAVIS [53] and AutoGeo [16]. UI data are obtained from the MultiUI dataset [24], while poster images feature OCR-based captions. Video captions are derived from OpenVid [32] and Panda [4], covering multiple attributes such as detailed descriptions, style, background, tags, camera angles, and object information. Fig. 6 illustrates the token length distribution for different caption types associated with natural images, categorized into detailed, medium, short, and tag captions.

B Experimental Setup

We fine-tune the Qwen2-VL-7B-Instruct model on a large-scale captioning dataset using 64 A100 GPUs. The training process is distributed using torchrun with the DeepSpeed ZeRO-3 optimization strategy.

Hyperparameters:

- **Batch Size:** 256 (1 per device, with gradient accumulation of 8)
- **Learning Rate:** 1e-5 (base model), 1e-5 (merger module), 2e-6 (vision tower)
- **Weight Decay:** 0.0
- **Warmup Ratio:** 3%
- **Scheduler:** Cosine decay
- **Precision:** BF16 enabled,
- **Gradient Checkpointing:** Enabled

Training Details:

- **Image Resolution:** From $2 \times 28 \times 28$ to $6400 \times 28 \times 28$ pixels
- **Epochs:** 1

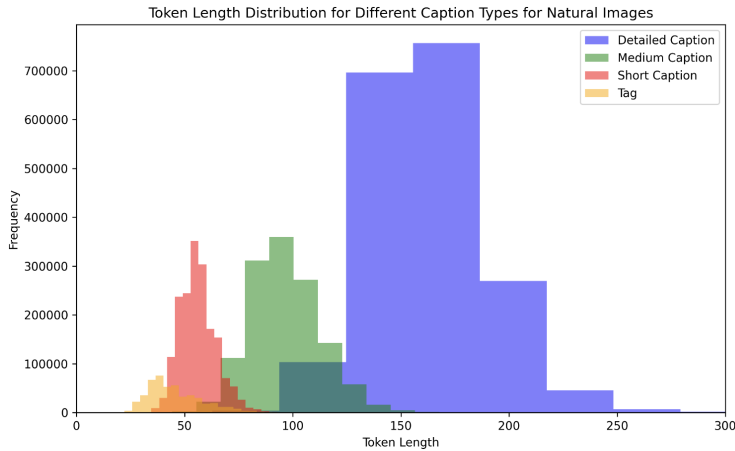


Figure 6: Token length distribution for natural images.

C System Prompt Example

Fig. 8 presents different system prompts used in OMNICATIONER for various image types. It categorizes prompts into three sections: natural images, visual text images, and structured images. These prompts guide the model’s captioning style and task-specific adaptations.

Natural Image (Data Source: In-house)		
Detailed_en (1.9M)	Detailed_zh (1.9M)	Medium_en (650K)
Medium_zh (642K)	Short_en (350K)	Short_zh (324K)
Tag_en (200K)	Tag_zh (169K)	
Nature Image (Data Source: BLIP3Kale)		
Detailed_en (55K)	Detailed_zh (132K)	Medium_en (483K)
Medium_zh (465K)	Short_en (1.6M)	Short_zh (1.3M)
Nature Image (Data Source: DenseFusion)		
Detailed_en (432K)	Detailed_zh (348K)	Medium_en (19K)
Medium_zh (13K)	Short_en (1K)	Short_zh (2k)
Table (Data Source: arXiv, MMtab)		
To LaTeX(2.8M)	Detailed_en (168K)	Detailed_zh (34K)
Chart (Data Source: arXiv, Tinychart)		
To Markdown (767K)	Detailed_en (253K)	Detailed_zh (79K)
Math-Equation (Data Source: arXiv)		
To LaTeX (3M)	Detailed_en (382K)	Detailed_zh (141K)
Math-Geometry (Data Source: Self-generated, Mavis, Autogeo)		
To LaTeX (102K)	Detailed_en (300K)	
Poster (Data Source: In-house)		
OCR (82K)	Detailed_en (134K)	Detailed_zh (98K)
PDF and UI (Data Source: MultiUI, arXiv)		
UI Caption (709K)	PDF OCR (47K)	Pure Text OCR (2M)
Video (Data Source: Openvid, Pandas)		
Detailed_en (600K)	Tag (600K)	Main Object (600K)
Style (600K)	Medium (600K)	Short (600K)
Background (600K)	Camera (600K)	

Figure 7: Dataset composition for pretraining OMNICATIONER.

D Caption Visualization

As illustrated in Fig. 9 to Fig. 15, we present a comprehensive visualization of captioning results across multiple tasks using OMNICATIONER, including natural images, table images, chart images, math images, poster images, and videos. For natural images, we demonstrate the impact of different system prompts on caption generation, showcasing how specific prompts can elicit world knowledge in the model’s responses in Fig. 16. In the case of structured images from Fig. 17, different system prompts lead to distinct stylistic variations in captioning, reflecting the adaptability of the model to various formatting requirements. Additionally, we visualize how OmniCaptioner-generated captions can enhance DeepSeek-R1-Distill-LLaMA-70B in Fig. 18, Fig. 19 and Fig. 20, enabling it to tackle visual tasks more effectively. These visualizations highlight the versatility and robustness of OMNICATIONER in handling diverse multimodal data, demonstrating its potential for improving vision-language understanding.

E Text-to-Image Generation

The visualization from Fig. 21 demonstrates that OMNICATIONER’s detailed captions significantly enhance the text-to-image (T2I) alignment in models like SANA 1.0 [43]. By providing precise and richly descriptive textual caption, the generated images exhibit improved fidelity to the original prompts. We also present some image conversion examples in Fig. 22 to illustrate the pixel-to-word ability of our OMNICATIONER. All the generated images shown above are produced by the generation model trained on image data labeled by OMNICATIONER, fine-tuned using SANA 1.0 with 1.6B parameters.

System Prompt For Natural Images:

"Detailed":

You are a helpful assistant focused on providing detailed descriptions and background information for the generated images. Analyze the given image and generate a comprehensive caption that includes the visual style, spatial relationships between elements, texture details, descriptions of the main objects, and relevant world knowledge to enhance understanding.

"Medium":

You are a helpful assistant specialized in creating medium-length captions for the generated images. Analyze the provided image and generate a caption that captures the key visual elements, while maintaining clarity and coherence.

"Short":

You are a helpful assistant focused on creating short captions for the generated images. Analyze the provided image and generate a concise caption that highlights the main subject.

"Tag":

You are a helpful assistant specialized in generating key tags for the generated images. Analyze the provided image and create a list of relevant tags that capture the main subjects, themes, and notable elements:

"Detailed_Zh":

你是一位专注于提供详细描述和背景信息的助手。分析给定的生成图像，生成一个全面的描述，包含视觉风格、元素之间的空间关系、纹理细节、主要对象的描述，以及增强理解的相关背景知识。

"Medium_Zh":

你是一位专注于创建中等长度图像描述的助手。分析所提供的生成图像，生成一个描述，捕捉关键视觉元素，保持清晰和连贯。

"Short_Zh":

你是一位专注于创建简短图像描述的助手。分析提供的生成图像，生成一个简洁的描述，突出主要主体。

"Tag_Zh":

你是一位专注于为图像生成关键词标签的助手。分析提供的生成图像，创建一个相关标签列表，捕捉主要主题、元素和显著特点。

"Detailed_Natural":

You are a helpful natural image captioner. Provide a comprehensive description of the natural image, including the main subject, background elements, lighting conditions, color distribution, textures, spatial arrangement, and any potential dynamic context.

"Medium_Natural":

You are a helpful natural image captioner. Describe the main content, background in the medium-length text.

"Short_Natural":

You are a helpful natural image captioner. Describe the main content, background in the short-length text.

"Detailed_Natural_Zh":

您是一位乐于助人的自然图片分析助手。请提供自然图像的全面描述，包括主要主题、背景元素、光照条件、颜色分布、纹理、空间排列以及任何潜在的背景。

"Medium_Natural_Zh":

您是一位乐于助人的自然图片分析助手。请用中等长度的文本描述主要内容和背景。

"Short_Natural_Zh":

您是一位乐于助人的自然图片分析助手。请用短文本描述主要内容和背景。

System Prompt For Visual Text Images:

"Visual_Text":

You are an advanced model designed to accurately analyze the image with text items. You can describe the text information and visual information in the image, including font style, size, color, background, text layout and other visual objects in detail.

"UI":

You are analyzing a UI webpage layout. Provide a detailed caption describing the layout's structure, including the arrangement, style, and functionality of key components such as buttons, navigation bars, input fields, and visual elements.

"OCR":

You are an advanced OCR model designed to accurately extract text from images. Your task is to analyze the provided image and return the text in a clear, readable format.

"Visual_Text_Zh":

你是一个精确分析文本内容图像的先进助手。你可以详细描述图像中的文本信息和视觉信息，包括字体样式、大小、颜色、背景、文本布局和其他视觉对象。

System Prompt For Structured Images:

"Chart":

You are a professional data visualization analyst. Given a chart image, first accurately perform OCR on any textual and numeric content (including titles, legends, axes, labels, and annotations), and you can convert it into Markdown format, then structure and analyze the extracted data to identify key trends and insights.

"Table":

You are a data conversion and extraction expert. Given a table image, you can convert it into CSV, HTML, Markdown or LaTeX formats, then extract and summarize the key relationships or insights from the data.

"Equation":

You are an equation analysis expert. Given an equation image, you can convert it into proper LaTeX format, then summarize any key mathematical properties, patterns, or insights it conveys.

"Geometry":

You are a geometry analysis expert. Given a geometric figure, you can convert it into a corresponding LaTeX representation, then provide insights or interpretations about the structure or properties.

"Structured_zh":

你是一个高级模型，旨在分析和解释图表、图形、表格、数学公式、数学几何图和数据可视化。你的任务是从提供的图像中提取相关信息，并以机器可读格式或结构化格式返回文本。

Figure 8: Different system prompts used for OMNICATIONER.



System Prompt Type: "Detailed"

Aerial photograph of a traditional Chinese temple complex perched on a rugged mountain landscape, likely during a festival. The central focus is a large, ornate pagoda with a golden roof, surrounded by smaller pagodas and pagoda-style structures, all intricately detailed with red and gold accents. Crowds of people are visible throughout the scene, moving along pathways and gathered around the buildings, indicating a bustling atmosphere. The terrain is rocky and partially covered with snow, with steep cliffs and narrow pathways leading to and around the structures. In the background, a series of distant mountains create a dramatic backdrop, with a soft blue hue from the sky. The lighting is warm and golden, suggesting a sunrise or sunset, casting long shadows and highlighting the textures of the rocks and the architectural details. High contrast, vibrant colors, dynamic composition, cultural richness, serene yet lively, visually stunning, panoramic perspective.

System Prompt Type: "Medium"

Aerial view of a traditional Chinese temple complex on a snowy mountain, featuring pagoda-style buildings with golden roofs and red accents. Crowds fill the temple grounds and surrounding pathways, with a large pagoda at the center. The scene includes rugged, snow-covered mountains and distant peaks under a clear blue sky, captured in vibrant colors with high contrast and natural lighting, creating a serene yet dynamic atmosphere.

System Prompt Type: "Short"

An aerial view of a bustling temple complex with pagoda-style buildings perched on rugged mountain peaks, surrounded by a snow-dusted landscape and bathed in warm, golden hour light, creating a serene yet dynamic scene.

System Prompt Type: "Tag"

Aerial view, mountainous landscape, temple complex, golden pagodas, rocky terrain, vibrant colors, cultural significance, dramatic perspective, natural lighting, high contrast, serene atmosphere, expansive scale, architectural photography.

System Prompt Type: "Detailed_zh"

空中俯瞰一座历史悠久的寺庙群，它坐落在崎岖的悬崖边上，四周是白雪皑皑的山脉。寺庙的建筑风格传统，屋顶铺着黄色瓦片，屋檐装饰华丽。主要建筑位于悬崖中央，两侧是通往山顶的楼梯。寺庙周围聚集了许多人，显示出这是一个热闹的场景。悬崖上覆盖着白雪，与寺庙的暖色调形成鲜明对比。远处的山脉在柔和的蓝色天空中逐渐消失，暗示着这是清晨或黄昏的光线。高角度视角，丰富的色彩搭配，自然与建筑元素的完美融合，宁静而又充满活力的氛围，清晰的细节，自然与人类建筑的和谐共存。

System Prompt Type: "Medium_zh"

空中俯瞰一座历史悠久的寺庙群，寺庙坐落在崎岖的悬崖上，四周被群山环绕。寺庙的屋顶是金色的，墙壁是红色的，寺庙通过一座桥与悬崖相连。一条蜿蜒的小径通向山顶的两个金色塔楼，而一群身着传统服饰的人们正朝着寺庙前进。白雪覆盖了地面，与岩石的土色调形成对比。这一幕捕捉了黄昏时分的宁静而壮丽的氛围，突出了寺庙的宏伟和自然景观的美丽。

System Prompt Type: "Short_zh"

空中俯瞰一座历史悠久的寺庙群，寺庙屋顶铺着黄色瓦片，建筑色彩斑斓，坐落在崎岖的悬崖上，四周环绕着白雪皑皑的山脉，一条蜿蜒的小径通向山顶，整个场景在金色时刻的光线下显得格外迷人。

System Prompt Type: "Tag_zh"

寺庙，悬崖，山脉，黄昏，空中视角，传统建筑，人群，鲜艳色彩，自然美景，宁静



System Prompt Type: "Detailed"

Urban street scene featuring the iconic Flatiron Building in Toronto, positioned centrally with its distinctive red brick facade and green roof. The building is flanked by modern skyscrapers, including a tall, glass skyscraper to the left and a dark, reflective building to the right. In the foreground, a few pedestrians cross the street, with one pushing a stroller on the right. The street is lined with orange traffic cones and signs, indicating construction or roadwork. The background includes a cloudy sky and a mix of architectural styles, from modern to classic. The scene is captured in natural lighting with a cool color palette, emphasizing the contrast between the historic and contemporary urban elements. Wide-angle perspective, high contrast, sharp focus, architectural photography, dynamic composition, urban exploration, visually balanced.

System Prompt Type: "Medium"

A wide-angle shot of Toronto's Flatiron Building, a historic red-brick structure with a green roof, stands out against modern skyscrapers. Pedestrians cross the street in front, while a black car and construction barriers are visible in the foreground. The overcast sky and cool color palette create a balanced, dynamic urban scene with a mix of historic and contemporary architecture.

System Prompt Type: "Short"

A wide-angle urban photograph captures Toronto's Flatiron Building, a historic red-brick structure with a green spire, surrounded by modern skyscrapers, with a few pedestrians and a black car in the foreground, set against an overcast sky. "Tag": "Flatiron Building, Toronto, urban landscape, historic architecture, modern skyscrapers, city street, overcast sky, construction, pedestrians, traffic cones, vibrant reds, cool tones, dynamic composition, architectural photography."

System Prompt Type: "Tag"

Flatiron Building, Toronto, urban landscape, historic architecture, modern skyscrapers, city street, overcast sky, construction, pedestrians, traffic cones, vibrant reds, cool tones, dynamic composition, architectural photography

System Prompt Type: "Detailed_zh"

城市街道场景，以一座历史悠久的红砖建筑为中心，建筑顶部有一个绿色的圆锥形塔楼，位于画面的右侧。这座建筑有许多窗户和黑色的消防逃生梯。在画面的左侧，一座现代的玻璃建筑与一座具有独特绿色尖顶的摩天大楼形成对比。在画面的右侧，一座现代砖砌建筑，窗户很大，与历史建筑相得益彰。街道上人来人往，有些人正在过马路，有些人站在人行道上。交通锥和路标被放置在街道交叉口。天空阴沉，给场景带来了一种沉闷的灰色调。城市摄影，中等景深，自然漫射光，冷色调，历史与现代建筑的对比，动态构图，平衡的框架，轻微的饱和，城市氛围。

System Prompt Type: "Medium_zh"

城市景观，以一座历史悠久的红砖平房建筑为中心，建筑顶部为绿色圆锥形，背景是现代摩天大楼。场景中包括行人、推着婴儿车的人、停放的汽车和施工围栏。天空阴沉。照片捕捉到了新旧建筑风格的融合，以及一种忧郁、高对比度的氛围，细节清晰。

System Prompt Type: "Short_zh"

多伦多的标志性建筑——奥斗大厦，矗立在现代摩天大楼之中，街道场景热闹非凡。行人络绎不绝，施工围栏随处可见。这一切都映衬在阴云密布的天空下，构成了一幅充满活力且对比强烈的都市画卷。

System Prompt Type: "Tag_zh"

"城市景观，平铁大厦，现代摩天大楼，建筑对比，阴天，城市摄影，历史与现代建筑"

Figure 9: Natural image captioning.

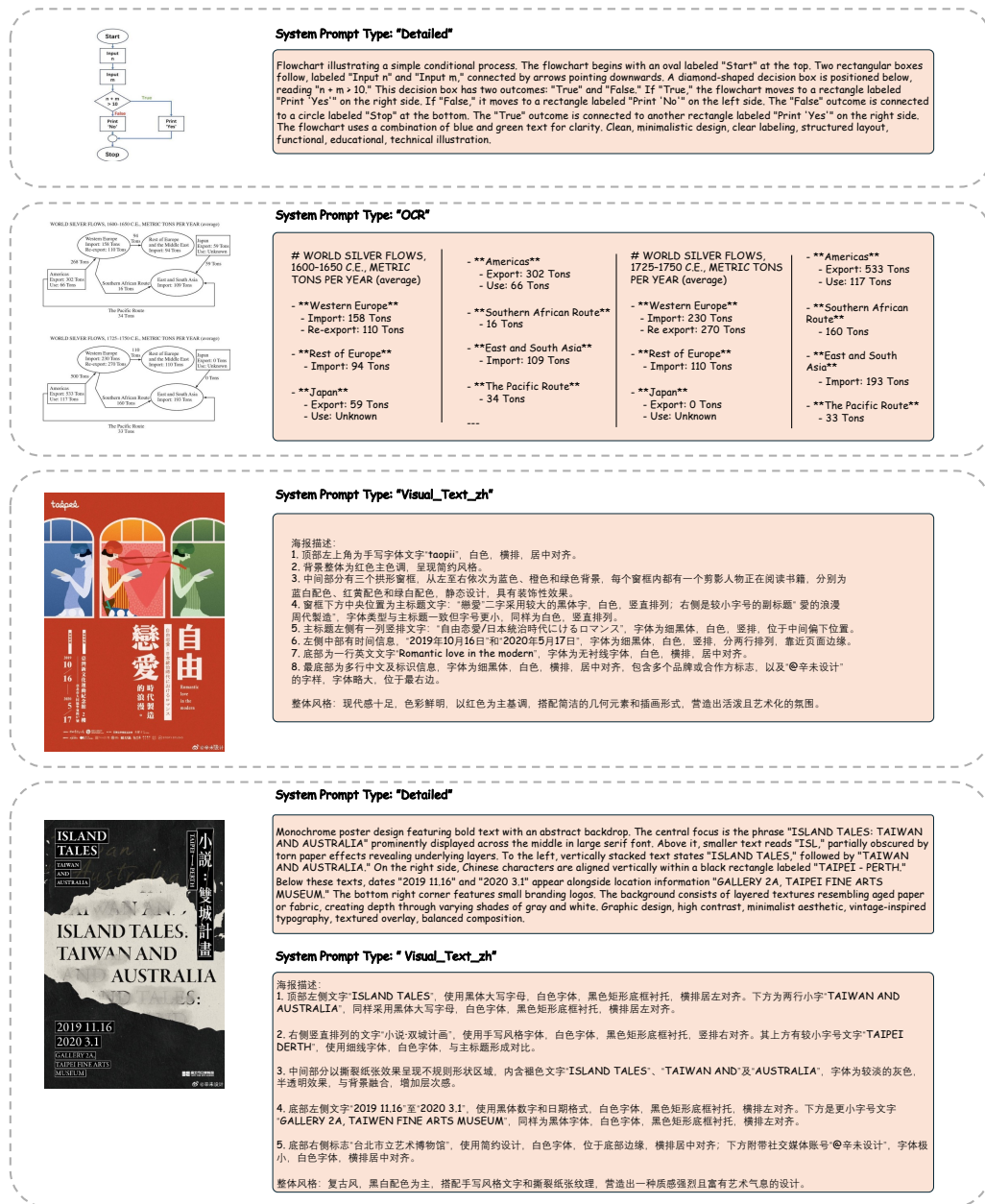
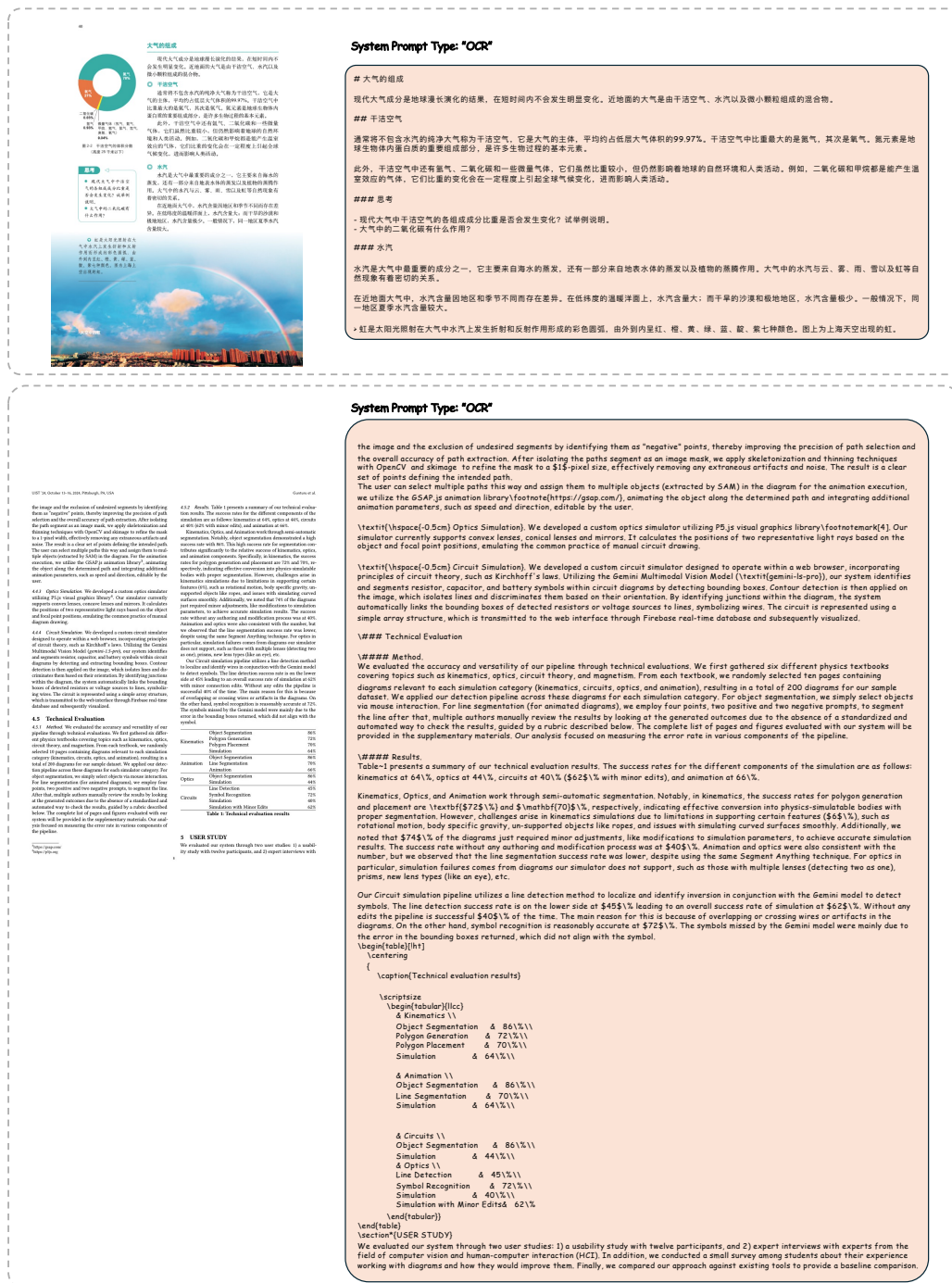


Figure 11: Visual-Text image captioning.



Figure 13: UI captioning.





System Prompt Type: "Video"

Short_Caption:

A serene cruise ship journey transitions into lively indoor activities onboard as passengers enjoy leisurely activities amidst vibrant surroundings.

Background_Caption:

The video begins at sea under soft daylight conditions, showcasing an expansive ocean view framed by distant landmasses. The ship's deck displays clean white structures against blue skies. Indoors, spacious rooms feature bright lighting, colorful decor, and modern furnishings, while communal areas host dynamic activities such as dining, music-making, and socializing beneath warm ambient lights.

Main_Object_Caption:

Passengers interact actively throughout various segments: one woman explores her cabin, adjusting items like bedding and storage spaces before transitioning to a lively lounge area where she engages in musical instruments. Her movements vary between deliberate organization indoors and expressive gestures during recreational activities. Her expressions shift subtly—calmness when tidying up and joy or focus during playing or conversing.

Reference_Caption:

The sequence opens outdoors with static medium shots capturing the sleek design of a massive cruise ship sailing across calm waters bathed in natural light. Transitioning inside, handheld close-ups follow individuals navigating their cabins adorned with personal touches like photos and art pieces. A woman organizes her space methodically, adjusting furniture and ensuring everything is in place. The camera shifts fluidly to showcase communal areas filled with activity. Vibrant reds and yellows dominate the interior tones, enhancing the festive mood as guests interact through music, dining, and conversation. Close-ups capture intricate details—the clinking of glasses over a table, fingers pressing drum pads rhythmically, and animated facial expressions among friends enjoying the evening's vibrancy. Static wide shots frame the entire setting for clarity and balance within compositions that highlight both individual actions and group dynamics. Subtle daily-like motions enhance intimacy without disrupting the scene's relaxed atmosphere. Lighting plays a critical role here: soft natural hues outside contrast sharply with warm artificial lighting indoors, creating visual harmony reflective of the blend between tranquil exploration and energetic camaraderie onboard.

Standard_Summary:

The video captures moments aboard a large cruise ship beginning with views of the vessel gliding across peaceful waters surrounded by distant landscapes. Inside, passengers engage in various leisure activities. One woman tidies her room attentively, organizing her belongings amid personal decorations. Later scenes depict lively communal areas with music-making, dining, and cheerful conversations unfolding dynamically. Warm lighting enhances these interactions, contrasting beautifully with the openness of outdoor views seen earlier. The smooth interplay between handheld close-ups, wide angles, and subtle tracking creates a balanced narrative of relaxation paired with spirited engagement onboard.

Style_Tags:

Travel Vlog, Cruise Ship Experience, Leisure Activities, Documentary-style Video, Lifestyle Clip

Key_Tags:

cruise ship, passengers, communal areas, daytime, relaxing environment, vibrant interiors, travel documentary style, handheld camera techniques, dynamic lighting contrasts, scenic ocean backdrop

Camera_Caption:

The frames exhibit varied shot types including Medium Shots, Close-Ups, and Wide Shots captured predominantly using eye-level perspectives. The progression suggests a combination of static framing and slight handheld motion, particularly evident in close-up sequences, likely due to minimal positional adjustments. Some frames display slow lateral movement resembling minor pans or tracking.



System Prompt Type: "Video"

Short_Caption:

A step-by-step DIY tutorial for creating decorative ceramic plates using paint pens on clay, followed by painting and assembling them into unique vases.

Background_Caption:

The setting is an indoor crafting area with wooden flooring, white walls adorned with shelves holding various tools and materials such as brushes, jars of paint, and containers filled with supplies. Natural light streams through large windows that reveal glimpses of greenery outside, suggesting it's daytime. The workspace remains static throughout but features scattered objects like trays containing clay pieces, jars labeled "Black," "White," or "Blue," and small bowls used to mix paints.

Main_Object_Caption:

The main focus shifts between two hands actively engaging in creative activities involving clay and painted ceramics. Initially, hands press a textured rolling pin over clay to create oval-shaped bases, which are later dried under sunlight before being painted. With steady movements, they use fine-tipped paint markers to draw intricate patterns resembling animal prints onto the clay surfaces while maintaining precision. After drying again, the painted pieces are carefully assembled together to form vases, secured firmly within their final structure. Each action emphasizes care and attention to detail during both preparation and decoration processes.

Reference_Caption:

Two hands diligently transform clay into artistic ceramic pieces starting at home-based workstations. Using a textured rolling pin, flat clay sheets are flattened evenly against a surface. The dried clay pieces are then cut into oval shapes suitable for decorating. Paint markers, pre-labeled in black, white, blue tones, are employed skillfully—creating elaborate patterns mimicking animal fur across several clay slabs. These decorated elements undergo further steps where individual pieces are aligned and attached to each other, forming functional yet stylish vases. In natural daylight pouring through nearby windows, the vibrant greens visible beyond add warmth and liveliness to the scene. The camera captures these transitions via close-up shots emphasizing texture details—the grainy rolling pin, smooth paint strokes, and matte finish after baking—while medium shots provide context for the broader crafting setup. Static framing ensures consistency without dynamic movement, enhancing clarity and instructional quality. This immersive process culminates when the finished products are displayed side by side alongside unpainted pieces, showcasing creativity blended with utility.

Standard_Summary:

The sequence showcases how handcrafted ceramic vases are made, beginning with shaping clay oval bases using a textured rolling pin. Once dried, detailed designs featuring animal-like patterns are meticulously drawn on top utilizing paint pens. The vases are later assembled by connecting multiple clay pieces. Set indoors amidst organized supplies and bathed in soft natural lighting, every stage—from rolling pins to intricate patterns—is captured clearly. The result is aesthetically pleasing, blending artistry with functionality as the completed vases take shape.

Style_Tags:

DIY Tutorial, Educational Video, Handcrafting Process, Step-by-Step Guide, Artistic Creation, Indoor Setting, Creative Demonstration, Instructional Content.

Key_Tags: Ceramic plates, paint pens, clay rolls, textured rolling pin, vases, crafting room, natural lighting, stationary camera angles, artistic patterns, assembly techniques.

Camera_Caption:

Predominantly handheld shots alternating between Medium Shots and Close-Ups focusing on specific objects held up towards the camera. Minimal camera motion suggests mostly static positioning with slight adjustments to maintain alignment.

Figure 15: Video captioning.



Figure 16: Natural image captioning with different system prompts.

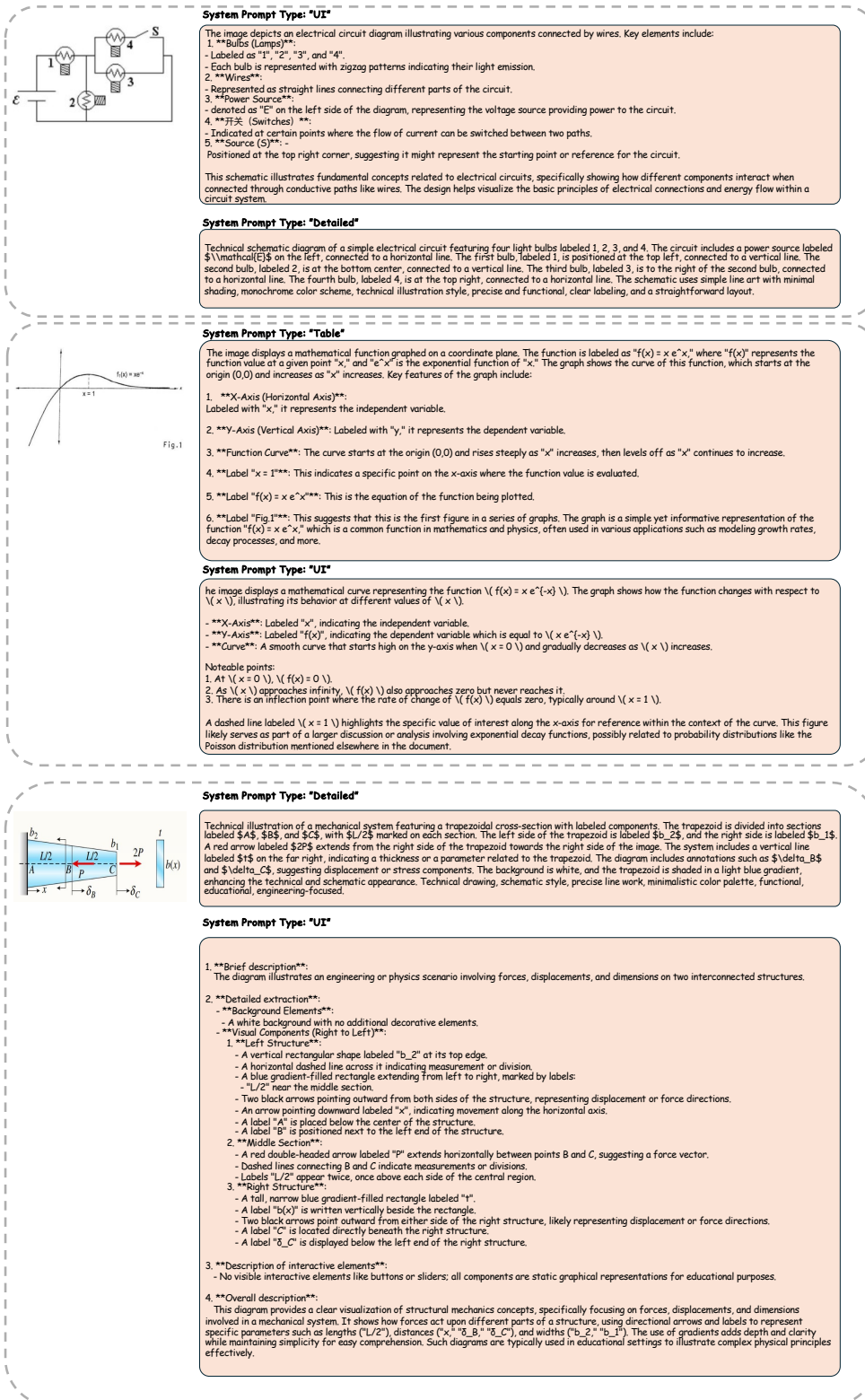


Figure 17: Structured image captioning with different system prompts.

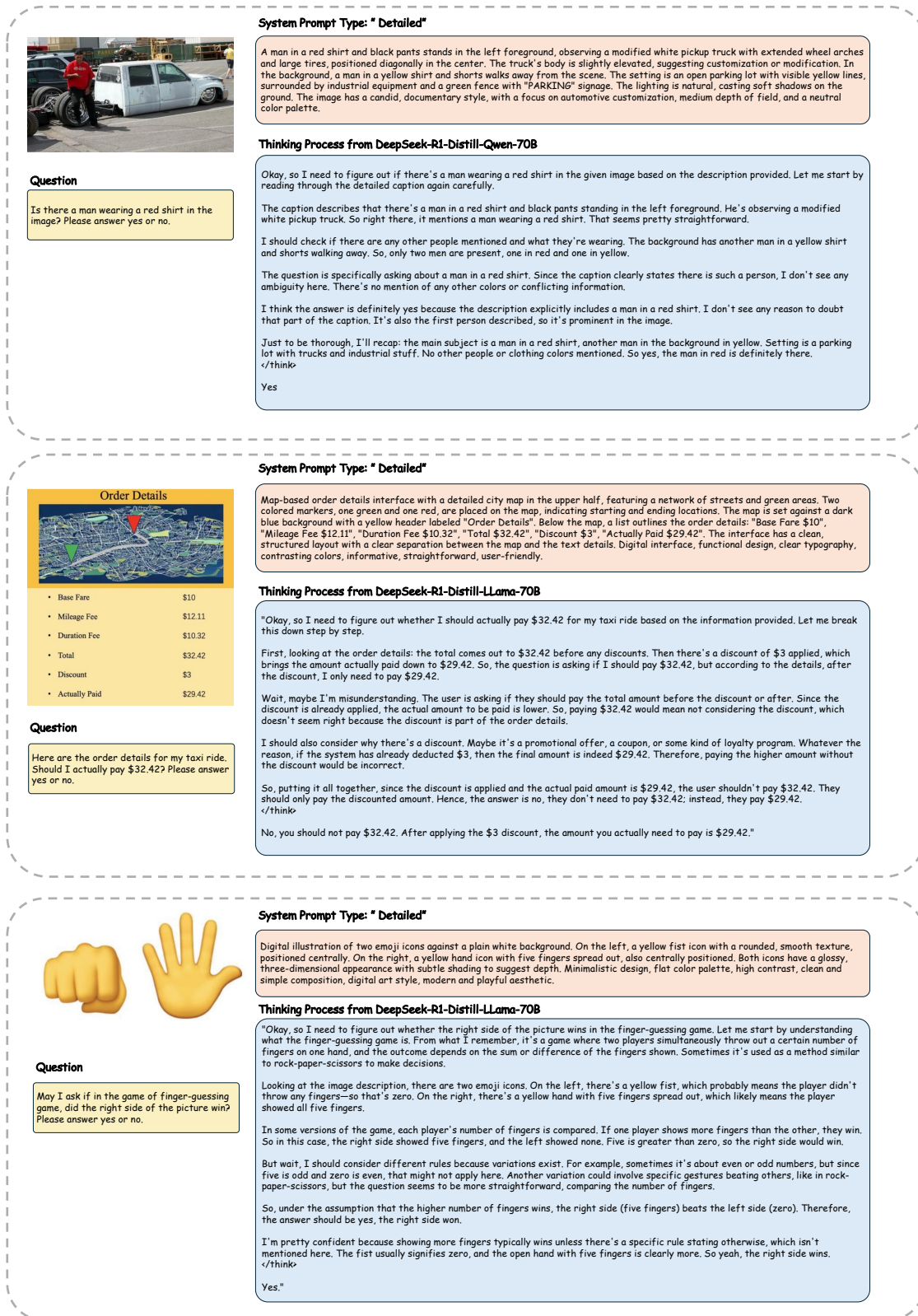
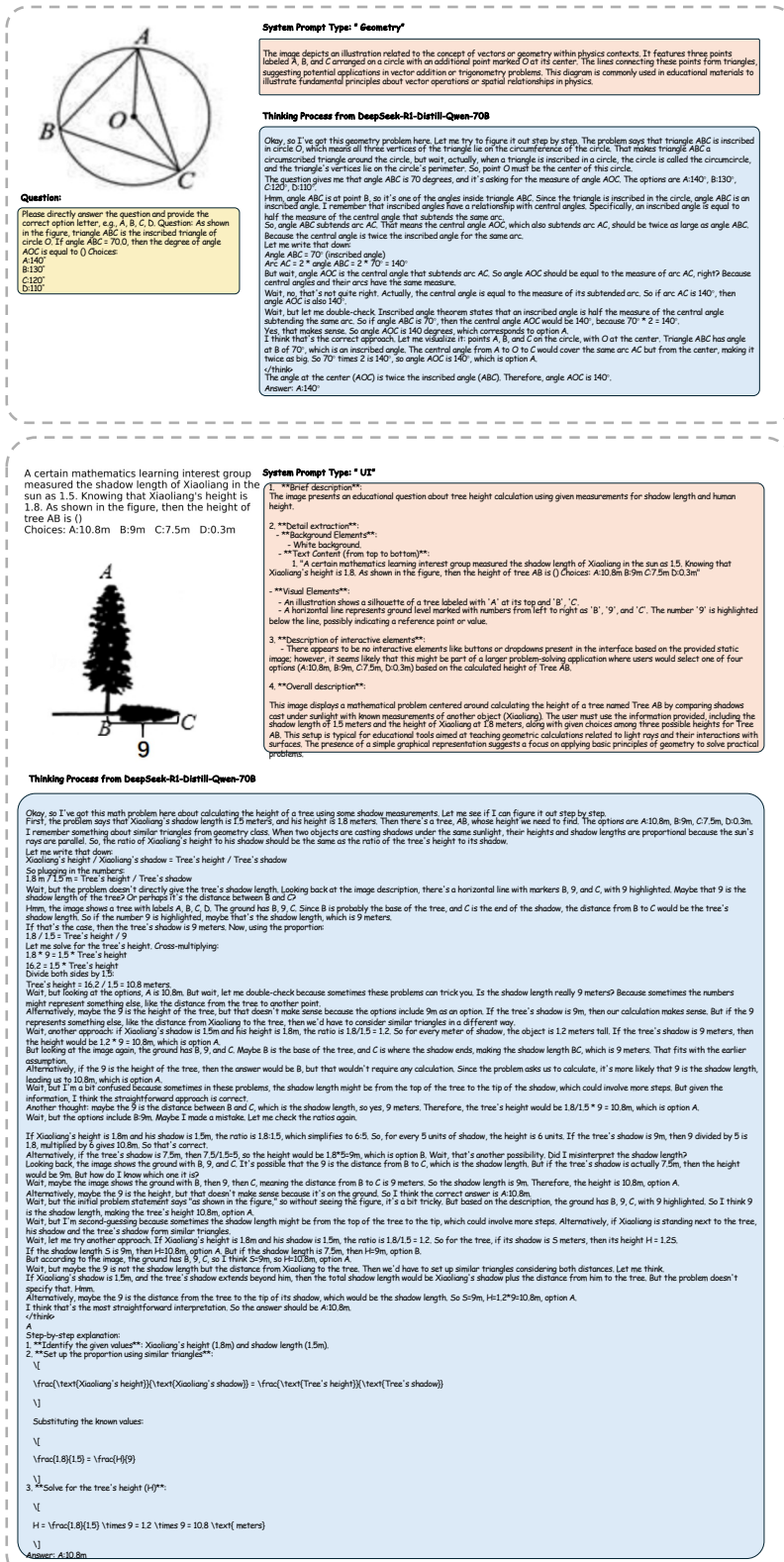


Figure 18: Visualization of thinking process with OMNICATIONER for natural images.



A certain mathematics learning interest group measured the shadow length of Xiaoliang in the sun as 1.5. Knowing that Xiaoliang's height is 1.8. As shown in the figure, then the height of tree AB is () Choices: A:10.8m B:9m C:7.5m D:0.3m

System Prompt Type: "UI"

1. **Intent description**:
The image presents an educational question about tree height calculation using given measurements for shadow length and human height.

2. **Detail extraction**:
- **Background Elements**:
- While background.
- **Main Content (from top to bottom)**:
1. "A certain mathematics learning interest group measured the shadow length of Xiaoliang in the sun as 1.5. Knowing that Xiaoliang's height is 1.8. As shown in the figure, then the height of tree AB is () Choices: A:10.8m B:9m C:7.5m D:0.3m"

- **Visual Elements**:
- An illustration shows a silhouette of a tree labeled with 'A' at its top and 'B' at its base.
- A horizontal line represents ground level marked with numbers from left to right as 'B', '9', and 'C'. The number '9' is highlighted below the line, possibly indicating a reference point or value.

3. **Description of interactive elements**:
There appears to be no interactive elements like buttons or dropdowns present in the interface based on the provided static image; however, it seems likely that this might be part of a larger problem-solving application where users would select one of four options (A:10.8m, B:9m, C:7.5m, D:0.3m) based on the calculated height of Tree AB.

4. **Overall description**:
This image displays a mathematical problem centered around calculating the height of a tree named Tree AB by comparing shadows cast under sunlight with known measurements of another object (Xiaoliang). The user must use the information provided, including the shadow length of 1.5 meters and the height of Xiaoliang at 1.8 meters, along with given choices among three possible heights for Tree AB. This setup is typical for educational tools aimed at teaching geometric calculations related to light rays and their interactions with surfaces. The presence of a simple graphical representation suggests a focus on applying basic principles of geometry to solve practical problems.

Thinking Process from DeepSeek-R1-Distill-Qwen-70B

Okay, so I've got this math problem here about calculating the height of a tree using some shadow measurements. Let me see if I can figure it out step by step. First, the problem says that Xiaoliang's shadow length is 1.5 meters, and his height is 1.8 meters. Then there's a tree, AB, whose height we need to find. The options are A:10.8m, B:9m, C:7.5m, D:0.3m. I remember something about similar triangles from geometry class. When two objects are casting shadows under the same sunlight, their heights and shadow lengths are proportional because the sun's rays are parallel. So, the ratio of Xiaoliang's height to his shadow should be the same as the ratio of the tree's height to its shadow. Let me write that down:

Xiaoliang's height / Xiaoliang's shadow = Tree's height / Tree's shadow

So plugging in the numbers:

1.8 m / 1.5 m = Tree's height / Tree's shadow

Wait, but the problem doesn't directly give the tree's shadow length. Looking back at the image description, there's a horizontal line with markers B, 9, and C, with 9 highlighted. Maybe that 9 is the shadow length of the tree? Or perhaps it's the distance between B and C. Hmm, the image shows a tree with base B and a person Xiaoliang with base C. The ground is marked with B, 9, and C. Since B is the base of the tree and C is the end of the shadow, the distance from B to C would be the tree's shadow length. So if the number 9 is highlighted, maybe that's the shadow length, which is 9 meters.

If that's the case, then the tree's shadow is 9 meters. Now, using the proportion:

1.8 / 1.5 = Tree's height / 9

Let me solve for the tree's height. Cross-multiplying:

1.8 * 9 = 1.5 * Tree's height

16.2 = 1.5 * Tree's height

Divide both sides by 1.5:

Tree's height = 16.2 / 1.5 = 10.8 meters

Wait, but looking at the options, A is 10.8m. But wait, let me double-check because sometimes these problems can trick you. Is the shadow length really 9 meters? Because sometimes the numbers might represent something else, like the distance from the tree to the tip of the shadow. Alternatively, maybe the 9 is the height of the tree, but that doesn't make sense because the options include 9m. If the tree's shadow is 9m, then our calculation makes sense. But if the 9 represents something else, like the distance from Xiaoliang to the tree, then we'd have to consider similar triangles in a different way.

Wait, another approach: if Xiaoliang's shadow is 1.5m and his height is 1.8m, the ratio is 1.8/1.5 = 1.2. So for every meter of shadow, the object is 1.2 meters tall. If the tree's shadow is 9 meters, then the height would be 1.2 * 9 = 10.8m, which is option A.

But looking at the image again, the ground has B, 9, and C. Maybe B is the base of the tree, and C is where the shadow ends, making the shadow length BC, which is 9 meters. That fits with the earlier assumption.

Alternatively, if the 9 is the height of the tree, then the answer would be B, but that wouldn't require any calculation. Since the problem asks us to calculate, it's more likely that 9 is the shadow length, leading us to 10.8m, which is option A.

Wait, but I'm a bit confused because sometimes in these problems, the shadow length might be from the top of the tree to the tip of the shadow, which could involve more steps. But given the information, I think the straightforward approach is correct.

Another thought: maybe the 9 is the distance between B and C, which is the shadow length, so yes, 9 meters. Therefore, the tree's height would be 1.8/1.5 * 9 = 10.8m, which is option A.

Wait, but the options include B:9m. Maybe I made a mistake. Let me check the ratios again.

If Xiaoliang's height is 1.8m and his shadow is 1.5m, the ratio is 1.8:1.5, which simplifies to 6:5. So, for every 5 units of shadow, the height is 6 units. If the tree's shadow is 9m, then 9m divided by 5 is 1.8, multiplied by 6 gives 10.8m. So that's correct.

Alternatively, if the tree's shadow is 7.5m, then 7.5/1.5 = 5, so the height would be 1.8*5 = 9m, which is option B. Wait, that's another possibility. Did I misinterpret the shadow length? Looking back, the image shows the ground with B, 9, and C. It's possible that the 9 is the distance from B to C, which is the shadow length. But if the tree's shadow is actually 7.5m, then the height would be 9m. But how do I know which one it is?

Wait, maybe the image shows the ground with B, 9, C, and the tree is at B. If the distance from B to C is 9m, then the shadow length is 9m. Therefore, the height is 10.8m, option A.

Alternatively, maybe the 9 is the height, but that doesn't make sense because it's on the ground. So I think the correct answer is A:10.8m.

Wait, but the initial problem statement says "as shown in the figure", so without seeing the figure, it's a bit tricky. But based on the description, the ground has B, 9, C, with 9 highlighted. So I think 9 is the shadow length, making the tree's height 10.8m, option A.

Wait, but I'm second-guessing because sometimes the shadow length might be from the top of the tree to the tip, which could involve more steps. Alternatively, if Xiaoliang is standing next to the tree, his shadow and the tree's shadow form similar triangles. Wait, let me try another approach. If Xiaoliang's height is 1.8m and his shadow is 1.5m, the ratio is 1.8/1.5 = 1.2. So for the tree, if its shadow is 5 meters, then its height H = 1.2 * 5 = 6m. If the shadow length is 9m, then H = 1.2 * 9 = 10.8m, option A. But if the shadow length is 7.5m, then H = 1.2 * 7.5 = 9m, option B.

But according to the image, the ground has B, 9, C, so I think it's 9m, so H = 10.8m, option A.

Wait, but maybe the 9 is not the shadow length but the distance from Xiaoliang to the tree. Then we'd have to set up similar triangles considering both distances. Let me think.

If Xiaoliang's shadow is 1.5m, and the tree's shadow extends beyond him, then the total shadow length would be Xiaoliang's shadow plus the distance from him to the tree. But the problem doesn't specify that. Hmm.

Alternatively, maybe the 9 is the distance from the tree to the tip of its shadow, which would be the shadow length. So 9m, H = 1.2 * 9 = 10.8m, option A.

I think that's the most straightforward interpretation. So the answer should be A:10.8m.

</think>

A

Step-by-step explanation:

1. **Identify the given values**:
Xiaoliang's height (1.8m) and shadow length (1.5m).

2. **Set up the proportion using similar triangles**:

$$\frac{\text{Xiaoliang's height}}{\text{Xiaoliang's shadow}} = \frac{\text{Tree's height}}{\text{Tree's shadow}}$$

Substituting the known values:

$$\frac{1.8}{1.5} = \frac{H}{9}$$

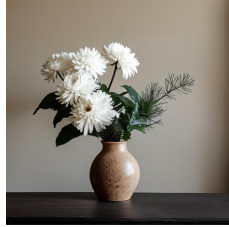
3. Solve for the tree's height (H):

$$H = \frac{1.8 \times 9}{1.5} = 1.2 \times 9 = 10.8 \text{ (meters)}$$

Answer: A:10.8m

Figure 20: Visualization of thinking process with OMNICATIONER for math images.

Minimalist still life photograph of a floral arrangement in a rustic ceramic vase, centrally positioned on a dark wooden surface. The vase is earthy brown with a textured finish. The arrangement features large, white chrysanthemums on the left, accompanied by glossy green leaves. On the right, there are delicate pine branches with needle-like leaves and a sprig of soft, feathery foliage. The background is a smooth, neutral beige wall, creating a serene and uncluttered atmosphere. Natural lighting, soft shadows, muted color palette, balanced composition, organic textures, tranquil, understated elegance.



Dramatic seascape photograph featuring a large, illuminated cumulus cloud dominating the center of the image against a deep blue sky. The cloud is brightly lit, creating a stark contrast with the dark ocean waves crashing onto the shore in the midground. The shoreline curves from the bottom left to the center, with wet sand reflecting the cloud's light. The horizon line is slightly above the center, with the ocean stretching out to the right. The lighting is intense and focused on the cloud, casting a soft glow on the wet sand. The scene is serene yet dramatic, with a moody atmosphere. Photorealistic, high contrast, vivid blue and white color palette, natural lighting, wide depth of field, tranquil yet powerful, visually balanced.



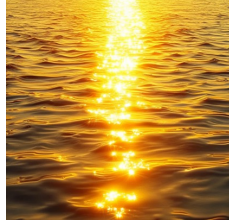
Close-up portrait of a young woman with long dark hair, looking directly at the camera. She is wearing a colorful, patterned sweater with shades of purple, green, and red. Her expression is neutral, with subtle makeup highlighting her eyes and lips. The background features a softly lit bedroom with a bed partially visible on the left, a lamp casting warm light, and a blurred piece of furniture or clothing in the background. The lighting is soft and natural, creating a warm and intimate atmosphere. The image has a shallow depth of field, focusing sharply on the woman's face while the background remains softly blurred. Photorealistic, warm color palette, intimate, serene, visually balanced.



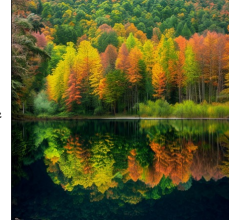
Illustration of a rusted, partially submerged military structure with a large gun turret in the right foreground, surrounded by lush greenery on top. A person stands on the structure's edge, gazing into the distance. The background features towering, fluffy white clouds against a clear blue sky. The sea occupies the lower half of the image, with gentle waves and a few small figures swimming in the distance. The structure's weathered orange and brown tones contrast with the vibrant blue and white of the sky and clouds. Stylized, semi-realistic, high contrast, vivid color palette, dynamic composition, serene atmosphere, illustrative art style, sense of isolation and exploration.



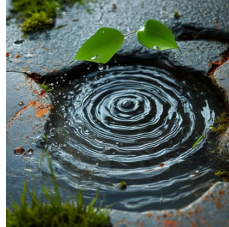
A serene photograph capturing the golden reflection of the sun on a vast expanse of water. The sun is positioned at the top center, casting a brilliant, shimmering trail of light across the rippling surface. The water is textured with gentle waves, creating a rhythmic pattern that leads the eye towards the horizon. The entire scene is bathed in warm, golden hues, enhancing the tranquil and meditative atmosphere. High contrast, natural lighting, golden hour, photorealistic, expansive composition, reflective surface, peaceful, visually harmonious.



Serene landscape photograph of a dense forest reflected in a still lake. The forest occupies the upper half of the image, with a variety of trees displaying autumn foliage in shades of green, yellow, and orange. The reflection in the lake below mirrors the forest perfectly, creating a symmetrical visual effect. The left side of the image shows a slightly denser cluster of trees, while the right side is more open, revealing more of the water's surface. The water is dark and still, enhancing the reflection. Natural lighting, high contrast, vibrant autumn colors, photorealistic, tranquil atmosphere, balanced composition, reflective symmetry, crisp detail, peaceful and contemplative mood.



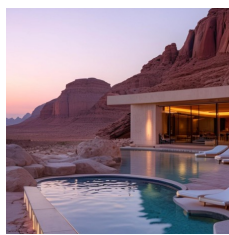
Close-up captures raindrops splashing onto the wet, textured surface of stone, creating concentric ripples. Two green leaves fall onto the water, one in the upper left corner and the other in the bottom center. Small patches of green moss and grass sprouts peek out from cracks in the stone, adding natural detail to the image. The stone takes on a dark, earthy hue with traces of rust and moss, contrasting with the bright green leaves. Natural light, high contrast, shallow depth of field, tranquil atmosphere, organic textures, dynamic water patterns, earthy color palette, tranquil, contemplative mood.



A close-up photograph of an otter standing in shallow, icy water, with a thick layer of snow resting on its head, resembling a hat. The otter is positioned slightly to the right of the center, facing forward with a curious expression. Its fur is a mix of dark brown and white, with the white fur more prominent on its face and chest. The background is a blurred expanse of icy water with patches of snow, creating a serene winter scene. Natural lighting, high contrast, photorealistic, sharp focus on the otter, soft focus on the background, cool color palette, tranquil, whimsical, visually balanced.



Modern architectural photography showing a luxury desert resort at dusk. In the foreground, a tranquil swimming pool is surrounded by rocky landforms, reflecting soft ambient light. Lounge chairs are placed around the pool, facing the water. In the midground, a modern-style building with large windows and warm interior lighting contrasts with the rugged desert landscape. In the background, towering rocky landforms appear particularly spectacular under the pastel sky. The entire scene is serene and luxurious, with natural and artificial elements blending harmoniously. Photography style: Medium Depth of Field, Soft Natural Light, Warm Color Pairing, High Contrast, Architectural Photography, Tranquility, Visual Balance, Harmonious Blend of Natural and Artificial Elements.



This is an illustration of a young man sitting in a traditional Japanese room. He is in the center of the frame, wearing a white shirt and green pants, holding a book with a red cover. The room is filled with various objects, including a vending machine filled with colorful drinks in the foreground and a green and orange fish-shaped ornament hanging above his head. The walls are decorated with posters and art, including a yellow bird above his head. There are several wooden chairs scattered around the room, one of which is draped with a blue towel. Sunlight shines through the windows, illuminating the scene and casting soft shadows. This illustration is colorful and clean-lined, with a manga style, warm color combinations, dynamic composition, and a warm, inviting atmosphere.



Figure 21: The detailed caption from OMNICATIONER enhances the alignment capability of text-to-image generation by providing precise descriptions, ensuring that the generated image accurately reflects the intended concepts, attributes, and relationships. The generation model here is fine-tuned on images labeled by OMNICATIONER, using the SANA 1.0 model with 1.6B parameters.

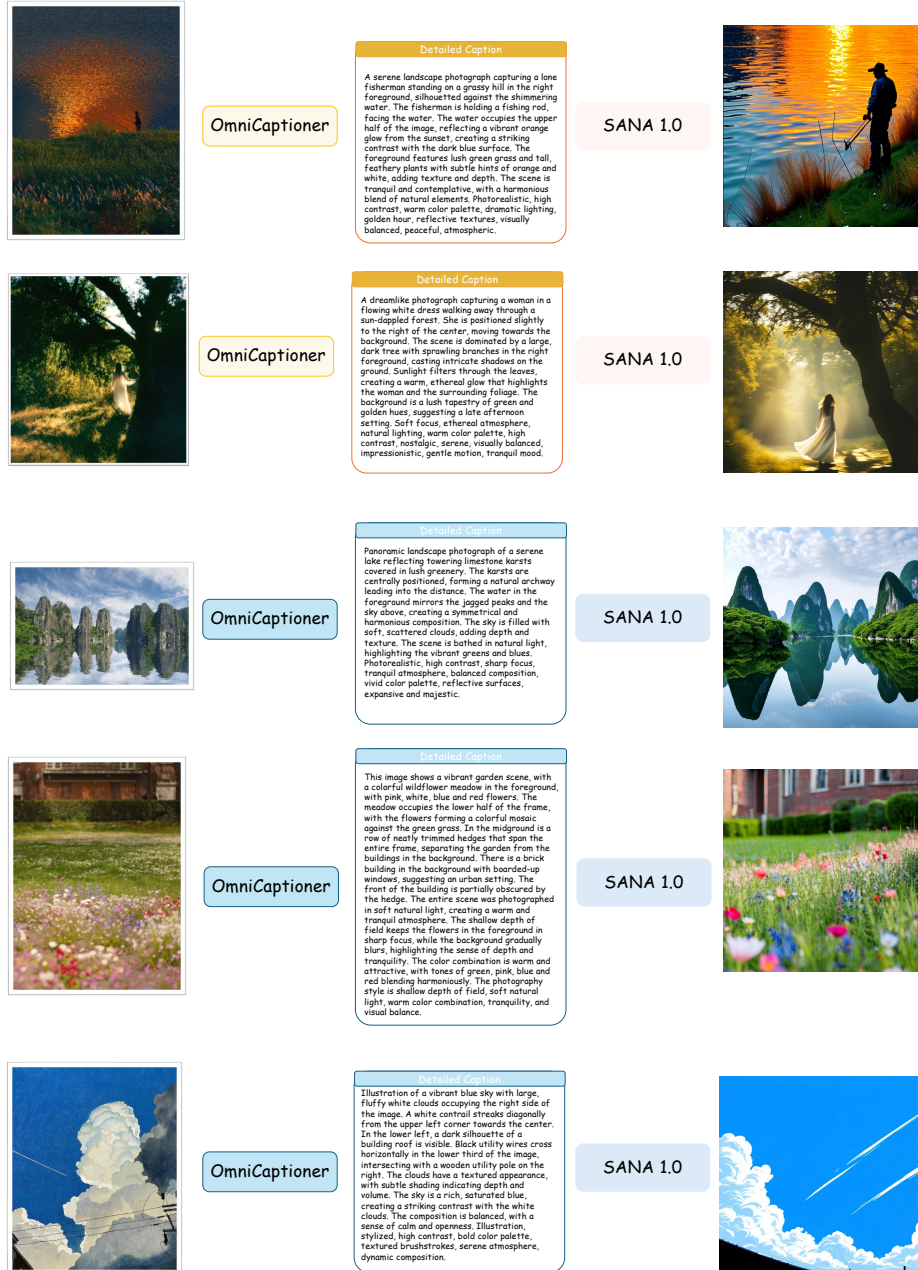


Figure 22: Image Conversion through OMNICATIONER and SANA-1.0. The generation model, SANA-1.0, is fine-tuned on images annotated by OMNICATIONER, enabling more accurate and semantically aligned image generation.